

Experience Sharing Studio: An Immersive Platform for Spatio-Temporal Recording and Playback of Embodied Multisensory Memory

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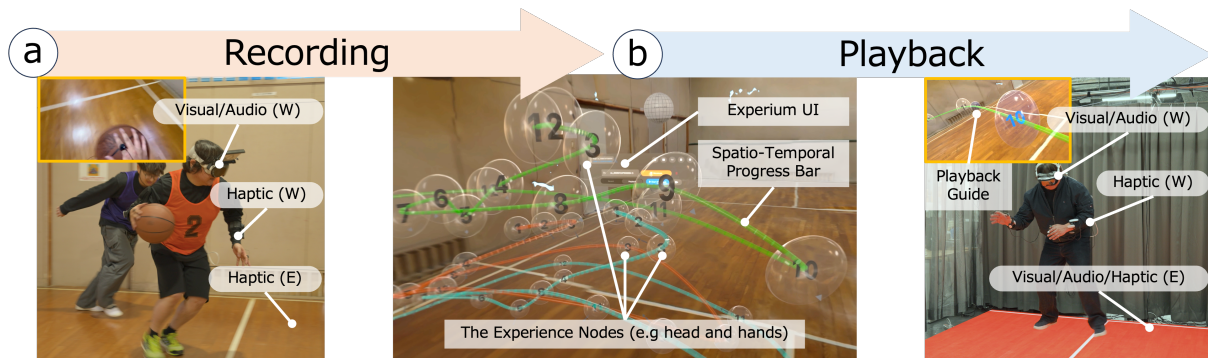


Figure 1: Overview of the Experience Sharing Studio (ESS) and Experium Platform. ESS integrates wearable spatial display, haptic devices, and floor modules. Orchestrated by Experium, the system enables spatio-temporal recording and playback of embodied multisensory memories. (a) Spatio-temporal recording of audiovisual-haptic experiences arranged as nodes. (b) Playback of recorded multisensory data within ESS. W: Wearable; E: Environmental.

Abstract

Conventional digital archives remain predominantly audiovisual, still lacking the haptic and embodied information crucial for sharing experiences with others. We propose the Experience Sharing Studio (ESS), an immersive platform for the spatio-temporal recording and playback of embodied multisensory memories. ESS and the Experium platform integrate multimodal data—including spatial



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video, haptics, and 3D body poses captured via wearable and environmental devices—into a unified format managed by a unique Experience ID. A key feature of this research is the interactive "re-living" capability: the platform synchronizes this integrated data to render a Spatio-Temporal Progress Bar that visualizes 3D body poses, guiding users through the playback of recorded experiences.

CCS Concepts

• **Human-centered computing** → **User interface management systems**.

Keywords

spatial computing, haptic feedback, experience sharing, multisensory memory, embodied interaction

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1 Introduction

Humans have long shared their experiences by recording them through visual and auditory media, such as photography and video. Recent advancements in haptic recording and playback [Tsujita et al. 2025; Umehara et al. 2024] have further enabled the synchronization and sharing of multisensory experiences. For instance, the Experience Sharing Box [Higashi et al. 2025] facilitated multisensory sharing within a confined, static environment. However, current research focuses primarily on recreating experiences in static settings; a platform for archiving wide-area, dynamic bodily experiences, including the grounding that supports their physical reality, has yet to be established.

Researchers have explored augmenting human skill acquisition via XR. A VR-based Kendama system [Kawasaki et al. 2022] assists beginners by visualizing success factors. While effective, such systems are task-specialized and do not serve as general-purpose platforms for multisensory memories. Furthermore, temporal manipulation approaches [Jiang and Rekimoto 2020] rely on predefined interventions during active training.

While these foundational approaches have provided significant insights, their scope has often been centered on temporal manipulation as a supportive intervention during active training. There remains an opportunity to further explore reliving interactions that allow users to intuitively navigate through past multisensory recordings—similar to scrubbing a video timeline—to synchronize their movements with expert body data in a highly immersive manner. In this paper, we propose the Experience Sharing Studio (ESS) to explore the potential of a platform for recording and playing back wide-area multisensory experiences involving dynamic physical movement with a strong sense of grounding and intuitive spatio-temporal navigation analogous to a video progress bar.

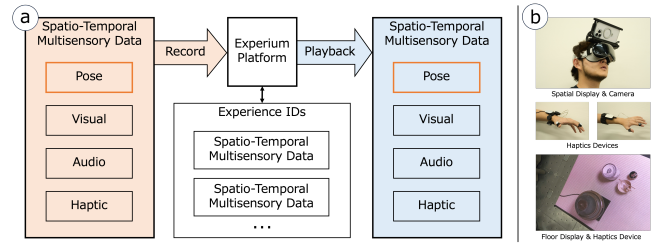


Figure 2: Data flow and device configuration of ESS. The platform links spatial display, haptic devices, and floor devices through Experience Data and Experience IDs for coordinated recording and playback.

2 Experience Sharing Studio

This section describes the core concepts and implementation of the ESS. Figure 2(a) illustrates the concept of spatio-temporal transitions of multisensory data within ESS, and Figure 2(b) shows the system's main components: the Spatial Display, Haptic Devices, and Floor Device.

2.1 Experium Platform and Experience Data

The Experium Platform acts as a software stack (e.g., visionOS application) that manages and stores integrated embodied multisensory data linked to a unique identifier called an Experience ID.

The platform handles "Experience Data," which consists of "Pose Data"—recording the 3D coordinates of the body—in addition to conventional audio-visual and haptic data. By loading an arbitrary Experience ID into the platform, users can spatio-temporally relive the physical experience of another person.

2.2 Spatio-Temporal Progress Bar

We define a "Spatio-Temporal Progress Bar" as a chronological sequence of "Experience Nodes," representing user Pose Data (e.g., head/hand 3D coordinates, palm directions) recorded during a session. Furthermore, when playing back Experience Data (Pose and audiovisual-haptics), a Playback Guide transitions along this Progress Bar, enabling intuitive 3D navigation.

2.3 Recording and Playback Environments

ESS utilizes wearable and environmental devices managed by the Experium Platform, which synchronizes network-based operations for all sessions.

- (1) **Wearable Devices:** These include Spatial Display (Apple Vision Pro) for pose tracking and rendering, Smartphone Camera (iPhone Pro) for spatial video, and Haptic Devices (SenseFuse/FeelFuse) for bidirectional haptic/force interaction.
- (2) **Environmental Devices:** Floor modules, vibration transducers, and LiDAR scanners support feedback, recording, and environmental modeling.

During playback, a Playback Guide transitions between Experience Nodes on the Spatial Progress Bar, synchronizing spatial

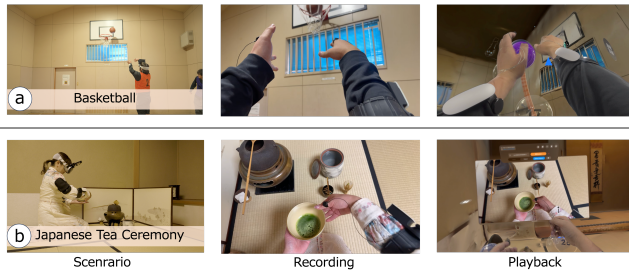


Figure 3: Use scenarios of ESS. (a) A dynamic basketball scenario demonstrates wide-area physical movement, grounding, and rhythm. (b) A Japanese tea ceremony scenario demonstrates subtle posture, hand motion, and spatial atmosphere.

video and haptics for each segment. Playback speed is fully adjustable; for instance, at $0.5\times$ speed, both the guide transition and the corresponding multisensory feedback are scaled proportionally to maintain synchronization.

2.4 User-Driven Playback Mode

The Experium Platform implements two playback modes tailored to user proficiency:

- (1) **Standard-Playback Mode (baseline)** A standard mode that replays Experience Nodes according to the original recorded timeline (e.g., 10 nodes recorded at 1-second intervals are presented at the same 1-second intervals).
- (2) **User-driven Playback Mode (Ours)** A mode where the Playback Guide's speed dynamically adjusts based on user performance. For instance, if the physical transition between nodes is slower than expected, the system automatically decelerates the guide (slow motion). Conversely, if the movement is accurate, it accelerates (closer to real-time). This temporal intervention supports efficient acquisition of complex embodied knowledge.

3 Use Scenarios

To demonstrate the versatility and expressive range of ESS as an integrated embodied multisensory experience sharing platform, we present two representative scenarios characterized by contrasting physical movement qualities.

3.1 Dynamic Scenario: Basketball

Figure 3(a) illustrates a basketball practice scene as a scenario involving wide-area and dynamic physical movements. This scenario targets a sequence of whole-body actions, ranging from the tactile feedback of the ball's impact during dribbling to the rapid shifts in the center of gravity required for feints and shooting. Specifically, through the coordination of floor vibration and wearable haptics, users can intuitively learn the physical rhythm and grounding—such as the sensation of kicking the floor—directly from the recorder's perspective.

3.2 Static Scenario: Japanese Tea Ceremony

Figure 3(b) presents the Japanese tea ceremony (Sadou) as a scenario requiring static, precise, and subtle gestures. Key elements here include the fluid wrist movements when using the Chasen (tea whisk) and the strict posture and timing maintained when serving the tea bowl to a guest.

Furthermore, the tatami texture projected onto the LED Floor and the recreation of the tranquil spatial atmosphere contribute not only to the reproduction of technical skills but also to sharing the experienter's context with external observers. This includes conveying non-verbal nuances such as "Ma" (the concept of negative space or pause) and the inherent spirituality of the scene.

4 Conclusion

In this paper, we proposed ESS, an immersive platform for the spatio-temporal recording and playback of embodied multisensory memory. We implemented a synchronized multimodal spatio-temporal recording and playback system. To mitigate the perspective discrepancy between recording and viewing, we designed and improved the Spatio-Temporal Progress Bar as an interactive playback interface. Furthermore, we considered the system's potential for diverse instructional applications, particularly through its capability to adaptively slow down playback in synchronization with the user's movements.

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